

# Intro to Laser Engraving (1/2)

## Design:

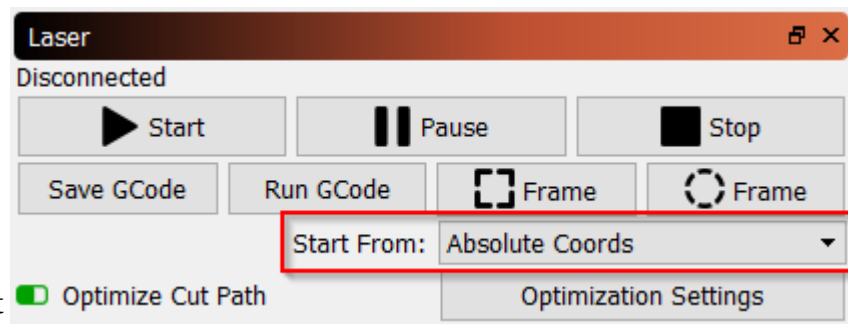
- Open the template tool (First QR code on reverse)
- Choose a template and font.
- Use the “Customize” tab to customize to your liking.
- Preview the design to show the SVG that will be used for cutting.
- Share Design to save it as an SVG.
- Upload the SVG to the file server for cutting:

## Prepare for cutting:

- Move over to Lightburn PC
- Open Firefox and go to <http://192.168.12.211/>
- Find your design in the uploads and download it to: Documents/Lightburn Projects/jarney
- Open Lightburn from the Desktop
- File → Import, open your design from the imports folder
- Select all and align your design to X,Y at (60mm,60mm) to line it up with the jig using the icons on the toolbar.
- Select Absolute Coordinates:
- Set the “Mode” and the “Speed/Power” for each of the colors
  - Images: Image mode, speed 250mm/s power 20%
  - Text (maroon): Fill or Line mode, speed 250mm/s power 20%
  - Lines (blue): Line mode, speed 250mm/s power 20%
  - Set the outermost line to turn off “Output”
- Confirm that “Cut Selected Graphics” is not turned on.
- Click “Send” button and give it a name (8 characters max)

## Perform the cut:

- Move over to laser cutter
- Place cutting board in jig, aligning it either “Portrait” or “Landscape” depending on your template.
- Use the “File” button and arrow keys on the laser cutter to select your file
- Press “Enter” to open up your file
- Press “Frame” button to draw the outline of your design and confirm it’s in the right place
- Confirm with your instructor that machine is ready to go (ventilation, chiller, power, air pump).



| #   | Layer | Mode  | Spd/Pwr      | Output | Show |
|-----|-------|-------|--------------|--------|------|
| C00 | 00    | Fill  | 50.0 / 100.0 |        |      |
| 10% | 01    | Fill  | 50.0 / 10.0  |        |      |
| 20% | 02    | Fill  | 50.0 / 20.0  |        |      |
| 30% | 03    | Fill  | 50.0 / 30.0  |        |      |
| 40% | 04    | Fill  | 50.0 / 40.0  |        |      |
| 50% | 05    | Fill  | 50.0 / 50.0  |        |      |
| 60% | 06    | Fill  | 50.0 / 60.0  |        |      |
| 70% | 07    | Fill  | 50.0 / 70.0  |        |      |
| 80% | 08    | Fill  | 50.0 / 80.0  |        |      |
| 90% | 09    | Fill  | 50.0 / 90.0  |        |      |
| 10% | 10    | Fill  | 50.0 / 100.0 |        |      |
| C29 | 29    | Image | 50.0 / 100.0 |        |      |
| C28 | 28    | Image | 50.0 / 100.0 |        |      |

Layer Color Speed (mm/s) 50.0  
Pass Count 1 Power Max (%) 100.00  
Interval (mm) 0.100 Power Min (%) 0.00

# Intro to Laser Engraving (2/2)

- Close door and press green “Start” button.
- Wait for machine to beep, do not leave unattended.

## Other References

Other references that may be useful for making your own designs

|                                                                                                                        |                                                                                                                               |                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <p>Spark Cutting Board Templates</p>  | <p>Location to upload files for cutting</p>  | <p>More Spark Studio Laser Resources</p>  |
| <p>Inkscape Tutorials</p>           | <p>Lightburn Tutorials</p>                 |                                                                                                                              |